State Variable Theory

#Chairing #Explanation

Given our ***Euler-Lagrange Formulation***:

$$B(q)\ddot{q}+C(\dot{q},q)\dot{q}+G(q)=\Omega\tau+\phi_{diss}(\dot{q},q,t)+\langle\lambda(t),f(q)\rangle$$

We can propose our **state variables**:

$$\begin{cases} q_a=q \\ q_b=\dot{q} \end{cases}$$

Then, the *dynamics* of the proposed **state variables** are:

$$\dot{q}_a=\dot{q}=q_b$$

\dot{q}_{b} = \ddot{q} = B^{-1}(q_{a}) [\Omega\tau +\phi_{diss}(q_{b},q_{a},t) + \langle \lambda(t

\begin{cases}
\dot{q}{a}=q{b} \
\dot{q}{b}= f(q{a},q{b}) + g(q{a})(\Omega\tau +\langle \lambda(t), f(q_{a}) \rangle)
\end{cases}

$Now, weneedtoproposeacontrollersowecancontrolourdynamicsdisregardingthe**system**, in$

\dot{q}{b} = f(q{a},q{b}) + g(q{a})(\Omega\tau +\langle \lambda(t), f(q{a}) \rangle) = - K{P}q{a} - K{D}q_{b}

$Todothis, wemaysimplyproposea***PDcontroller***forthe**torque**\$\tau\$thattakesintoa$

\tau=\Omega^{-1}[g^{-1}(q{a})(- K{P}q{a} - K{D}q{b} - f(q{a},q_{b}))] - \langle \lambda(t),f(q_{a}) \rangle

\tau=(J^T)^{-1}[B^{-1}(q{a})(- K{P}q{a} - K{D}q{b} - f(q{a},q_{b}))]

$$Whensubstitutingintothedynamicsweget:$$

\dot{q}{b} = f(q{a},q{b}) + g(q{a})(\cancel{ \Omega(\Omega^{-1} }[g^{-1}(q{a})(- K{P}q{a} - K{D}q{b} - f(q{a},q_{b}))] \cancel{ - \langle \lambda(t),f(q{a}) \rangle) +\langle \lambda(t), f(q{a}) \rangle })

\dot{q}{b} = f(q{a},q{b}) + \cancel{ g(q{a})[g^{-1}(q{a}) }(- K{P}q{a} - K{D}q{b} - f(q{a},q_{b}))]

\dot{q}{b} = \cancel{ f(q{a},q{b}) - f(q{a},q{b}) } - K{P}q{a} - K{D}q_{b}

\dot{q}{b} = -K{P}q{a} - K{D}q_{b}

Webasicallymadea ∗ ∗ ∗ controller ∗ ∗ ∗ forthe ∗ ∗system′sacceleration ∗ ∗byproposingoneforth

\begin{cases}
\dot{q}{a}=q{b} \
\dot{q}{b}=-K{P}q{a} - K{D}q_{b}
\end{cases}

In ∗ ∗vectorform ∗ ∗ :

\frac{d}{dt} \mathrm{q}=\frac{d}{dt} \begin{bmatrix} q{a} \ q{b} \end{bmatrix} =
\begin{bmatrix}
q{b} \
-K{P}q{a} - K{D}q_{b}
\end{bmatrix}

= \begin{bmatrix}
0{n\times n} & I{n\times n} \
-K{P} & -K{D}
\end{bmatrix} \begin{bmatrix}
q{a} \
q{b}
\end{bmatrix}

\frac{d}{dt} \mathrm{q} = A\mathrm{q}

From this, is obvious to note that q is a 12×1 vector, and both K_P and K_D ar

\begin{cases}
\dot{q}{a}=q{b} \
\dot{q}{b}= f(q{a},q{b}) + g(q{a})(\Omega\tau +\langle \lambda(t), f(q{a}) \rangle) + \psi(q{a},q_{b},t)
\end{cases}

$Where:$$ $\psi(q_a, q_b, .)$: Non-modelled sections of the robot.$$ $\psi(., ., t)$: Effect of all external pert

\Psi={ \psi:Q{a} \times TQ{a} \times \mathrm{R}^+ \to \mathrm{R}^n | \text{ } \lvert\lvert \psi \rvert\rvert^2 \leq \psi{0} + \psi{1}\lvert\lvert q{a} \rvert\rvert^2 + \psi{2} \lvert\lvert q_{b} \rvert\rvert^2}

$Where:$$ $Q_a \subset \mathrm{R}^n$: Configuration space (positions of q_a)$$ TQ_a : Tangent bundle of Q_a, represe

\tau=\Omega^{-1}[g^{-1}(q{a})(- K{P}q{a} - K{D}q{b} - f(q{a},q_{b}))] - \langle \lambda(t),f(q_{a}) \rangle

$$Soweendupwith:$$

\dot{q}{b} = -K{P}q{a} - K{D}q{b}+\psi(q{a},q_{b},t)

\frac{d}{dt} q = \underbrace{ \begin{bmatrix}
0{n\times n} & I{n\times n} \
-K{P} & -K{D}
\end{bmatrix} }{ A } \begin{bmatrix}
q{a} \
q{b}
\end{bmatrix} + \underbrace{ \begin{bmatrix} 0 \ I \end{bmatrix} }{ B } \psi

$$Wherewesaidourperturbations\$\psi\$areboundedby:$$

\lvert\lvert \psi \rvert\rvert^2 \leq \psi{0} + \psi{1}\lvert\lvert q{a} \rvert\rvert^2 + \psi{2} \lvert\lvert q_{b} \rvert\rvert^2